

Introduction to Python Programming

Python is a high-level, general-purpose programming language. It is widely used for software development, automation, web development, data analysis, artificial intelligence, scientific computing, and education.

Python is also one of the easiest programming languages for beginners to learn because its syntax is relatively simple and readable.

In this module, we will learn the basic building blocks of Python programming:

- What programming is
- What Python is
- Writing and running a Python program
- The `print()` function
- Comments
- Variables
- Basic data types
- Arithmetic operators
- Taking input from the user
- Converting input into numbers
- Creating simple programs

What is Programming?

A computer does not automatically know what we want it to do. We have to give it instructions.

Programming is the process of writing instructions that a computer can execute.

For example, if we want the computer to display the message:

Hello, World!

we can give Python the following instruction:

```
1 print("Hello, World!")
```

When Python executes this instruction, the computer displays:

```
1 Hello, World!
```

The instructions written by a programmer are called **source code**.

What is Python?

Python is a programming language. It allows us to write instructions in a form that is much easier for humans to read than machine-level instructions.

Python was created by **Guido van Rossum** and was first released in 1991.

Python is used in many areas, including:

- Web development
- Software development
- Data analysis
- Artificial intelligence
- Machine learning
- Automation
- Scientific computing
- Education

Important Note: Python is not the same thing as a text editor. Python is the programming language and interpreter, while an application such as Thonny provides an environment in which we can write and run Python programs easily.

Your First Python Program

Open the Python editor installed on your computer.

Type the following:

```
1 print("Hello, World!")
```

Run the program.

The output should be:

```
1 Hello, World!
```

This is a complete Python program.

Understanding print()

`print()` is a built-in Python function used to display information on the screen.

For example:

```
1 print("Python")
2 print("I am learning programming.")
3 print("This is my first Python program.")
```

Output:

```
1 Python
2 I am learning programming.
3 This is my first Python program.
```

Each `print()` statement displays its content on a separate line.

Printing Numbers

The `print()` function can also display numbers.

```
1 print(10)
2 print(25)
3 print(100)
```

Output:

```
1 10
2 25
3 100
```

Notice that numbers do not need quotation marks.

Compare:

```
1 print(25)
2 print("25")
```

Both display:

```
1 25
2 25
```

However, they are different types of values. The first is a number, while the second is text.

Printing Calculations

Python can perform calculations inside `print()`.

```
1 print(10 + 5)
2 print(20 - 8)
3 print(6 * 4)
4 print(20 / 5)
```

Output:

```
1 15
2 12
3 24
4 4.0
```

Python performs the calculation and then displays the result.

Comments

A comment is a note written in a program for humans.

Python ignores comments when it executes the program.

A comment begins with the `#` symbol.

For example:

```
1 # This program displays a greeting
2
3 print("Hello")
```

The comment does not appear in the output.

Comments can also be written after a statement:

```
1 print("Hello") # Display a greeting
```

Comments are useful for explaining code and making programs easier to understand.

Tip: Do not write unnecessary comments for obvious statements. Use comments when they provide useful information.

Basic Arithmetic Operators

Python can perform the basic mathematical operations that we commonly use.

Operator	Operation	Example
+	Addition	10 + 5
-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5
%	Remainder	10 % 3
**	Power	2 ** 3

Addition

```
1 print(15 + 7)
```

Output:

```
1 22
```

Subtraction

```
1 print(15 - 7)
```

Output:

```
1 8
```

Multiplication

```
1 print(15 * 7)
```

Output:

```
1 105
```

Division

```
1 print(15 / 5)
```

Output:

```
1 3.0
```

Python uses / for ordinary division. The result is normally a floating-point number.

Remainder

The % operator gives the remainder after division.

For example:

```
1 print(10 % 3)
```

Output:

```
1 1
```

This is because 10 divided by 3 gives a remainder of 1.

Another example:

```
1 print(20 % 4)
```

Output:

```
1 0
```

There is no remainder when 20 is divided by 4.

Power

The ** operator is used for powers.

```
1 print(2 ** 3)
2 print(5 ** 2)
```

Output:

```
1 8
2 25
```

Therefore:

$$2^3 = 8$$

and

$$5^2 = 25$$

Variables

A variable is a name used to store a value.

For example:

```
1 name = "Rahul"
2 age = 20
```

Here:

- `name` is a variable.
- `"Rahul"` is the value stored in `name`.
- `age` is a variable.
- `20` is the value stored in `age`.

We can display the values:

```
1 name = "Rahul"
2 age = 20
3
4 print(name)
5 print(age)
```

Output:

```
1 Rahul
2 20
```

Changing a Variable

A variable can be assigned a new value.

```
1 age = 20
2 print(age)
3
4 age = 21
5 print(age)
```

Output:

```
1 20
2 21
```

The value stored in `age` was changed from 20 to 21.

Variables Can Store Calculations

Variables can also store the result of calculations.

```
1 price = 100
2 quantity = 3
3
4 total = price * quantity
5
6 print(total)
```

Output:

```
1 300
```

This is much more useful than simply writing the number 300 because the program can calculate the result automatically when the values change.

Variable Naming Rules

Python has rules for variable names.

A variable name:

- Can contain letters.
- Can contain numbers.

- Can contain underscores.
- Cannot begin with a number.
- Cannot contain spaces.
- Is case-sensitive.

Valid examples:

```
1 name = "Rahul"
2 age = 20
3 student_name = "Rahul"
4 marks1 = 85
5 total_marks = 250
```

Invalid examples:

```
1 1name = "Rahul"
2 student name = "Rahul"
```

Python is case-sensitive.

Therefore:

```
1 name = "Rahul"
2 Name = "Amit"
```

creates two different variables.

Tip: Use meaningful variable names.

For example:

```
1 student_name = "Rahul"
2 student_age = 20
```

is easier to understand than:

```
1 x = "Rahul"
2 y = 20
```

Basic Data Types

A value stored in Python has a data type.

Four important beginner-level data types are:

Type	Name	Example
str	String	"Hello"
int	Integer	25
float	Decimal number	25.5
bool	Boolean	True

String

A string represents text.

```
1 name = "Rahul"
2 city = "Guwahati"
3 course = "Python"
```

Strings are normally written inside quotation marks.

Integer

An integer is a whole number without a decimal part.

```
1 age = 30
2 marks = 85
3 quantity = 10
```

Float

A float is a number that contains a decimal point.

```
1 height = 6.2
2 price = 99.50
3 temperature = 32.5
```

Boolean

A Boolean value can be either `True` or `False`.

```
1 is_student = True
2 is_raining = False
```

Boolean values will become particularly important when we learn conditions and decision-making.

Checking the Data Type

Python provides the `type()` function.

For example:

```
1 age = 30
2 print(type(age))
```

Output:

```
1 <class 'int'>
```

Another example:

```
1 name = "Rahul"
2 print(type(name))
```

Output:

```
1 <class 'str'>
```

Try:

```
1 print(type(25))
2 print(type(25.5))
3 print(type("Python"))
4 print(type(True))
```

Taking Input from the User

So far, our programs have used values that we entered directly into the program.

A useful program should also be able to receive information from the person using it.

Python provides the `input()` function for this purpose.

Example:

```
1 name = input("Enter your name: ")
2
3 print(name)
```

If the user enters:

```
1 Rahul
```

the program displays:

```
1 Rahul
```

We can also use the input inside a larger message:

```
1 name = input("Enter your name: ")
2
3 print("Hello", name)
```

If the user enters **Rahul**, the output will be:

```
1 Hello Rahul
```

Input and Variables

The value entered by the user can be stored in a variable.

```
1 city = input("Enter your city: ")
2
3 print("You live in", city)
```

The program does not know the city in advance. It receives the information when the user runs the program.

An Important Rule About `input()`

The `input()` function returns the user's input as a **string**.

For example:

```
1 age = input("Enter your age: ")
2
3 print(type(age))
```

If the user enters:

```
1 20
```

the output will still be:

```
1 <class 'str'>
```

This is important because we cannot directly use this input as an integer for mathematical calculations.

Consider:

```
1 num1 = input("Enter first number: ")
2 num2 = input("Enter second number: ")
3
4 print(num1 + num2)
```

If the user enters 10 and 20, the result will be:

```
1 1020
```

Python has treated the values as text and joined them together.

Converting Data Types

We can convert a value from one data type to another.

The most useful conversions for beginners are:

Function	Purpose
<code>int()</code>	Convert to an integer
<code>float()</code>	Convert to a decimal number
<code>str()</code>	Convert to text

Converting Input to an Integer

Instead of:

```
1 age = input("Enter your age: ")
```

we can write:

```
1 age = int(input("Enter your age: "))
```

Now the value stored in `age` is an integer.

For example:

```
1 num1 = int(input("Enter first number: "))
2 num2 = int(input("Enter second number: "))
```

```
3  
4 print(num1 + num2)
```

If the user enters 10 and 20, the output will be:

```
1 30
```

Converting Input to a Float

If the user needs to enter a decimal number, use `float()`.

```
1 price = float(input("Enter the price: "))  
2  
3 print(price)
```

The user can enter:

```
1 99.50
```

and Python will store it as a floating-point number.

Combining Text

Strings can be combined using the `+` operator.

For example:

```
1 first_name = "Rahul"  
2 last_name = "Sharma"  
3  
4 full_name = first_name + " " + last_name  
5  
6 print(full_name)
```

Output:

```
1 Rahul Sharma
```

The quotation marks containing a single space are used to place a space between the first name and last name.

Another simple method is to use commas with `print()`:

```
1 name = "Rahul"  
2 age = 20
```

```
3
4 print("Name:", name)
5 print("Age:", age)
```

Output:

```
1 Name: Rahul
2 Age: 20
```

Creating a Complete Program

Let us combine the concepts learned so far.

The following program asks for a student's name, age, and city.

```
1 name = input("Enter your name: ")
2 age = int(input("Enter your age: "))
3 city = input("Enter your city: ")
4
5 print("Name:", name)
6 print("Age:", age)
7 print("City:", city)
```

We can also calculate the student's age next year:

```
1 name = input("Enter your name: ")
2 age = int(input("Enter your age: "))
3 city = input("Enter your city: ")
4
5 print("Name:", name)
6 print("Age:", age)
7 print("City:", city)
8 print("Next year you will be", age + 1)
```

Practice Exercises

Write each program yourself. Do not simply copy the examples from the previous sections.

Exercise 1: Personal Information

Write a program that prints your:

- Name
- Age
- City
- Course

Use separate `print()` statements.

Exercise 2: Arithmetic

Write a program that displays the results of:

- $25 + 15$
- $50 - 17$
- 12×8
- $100 \div 4$

Exercise 3: Variables

Create two variables:

```
1 a = 25
2 b = 10
```

Display:

- The sum
- The difference
- The product
- The quotient

Exercise 4: User Name

Ask the user to enter their name.

Display:

```
1 Hello <name>
2 Welcome to Python programming.
```


Exercise 5: Age Calculator

Ask the user to enter their age.

Display their age next year.

For example:

```
1 Enter your age: 20
2 Next year you will be 21
```

Exercise 6: Rectangle Area

Ask the user to enter the length and width of a rectangle.

Calculate and display its area.

Use:

$$\text{Area} = \text{Length} \times \text{Width}$$

Exercise 7: Shopping Total

Ask the user to enter:

- Price of an item
- Quantity purchased

Calculate and display the total cost.

Use:

$$\text{Total Cost} = \text{Price} \times \text{Quantity}$$

Exercise 8: Square of a Number

Ask the user to enter a number.

Display its square.

For example:

```
1 Enter a number: 5
2 Square: 25
```

Exercise 9: Celsius to Fahrenheit

Ask the user to enter a temperature in Celsius.

Convert it to Fahrenheit using:

$$F = \frac{9}{5}C + 32$$

Display the result.

Exercise 10: Three Subject Marks

Ask the user to enter marks in three subjects.

Calculate and display:

- Total marks
- Average marks

Use:

$$\text{Total} = M_1 + M_2 + M_3$$

$$\text{Average} = \frac{\text{Total}}{3}$$

Mini Project: Student Information Program

Create a program that asks the user for:

1. Student name
2. Age
3. Course name
4. Marks in three subjects

The program should then display:

- Student name
- Age

- Course
- Total marks
- Average marks

A sample run should look similar to:

```
1 Enter student name: Rahul
2 Enter age: 20
3 Enter course: Python
4 Enter marks in Subject 1: 75
5 Enter marks in Subject 2: 82
6 Enter marks in Subject 3: 68
7
8 Student Name: Rahul
9 Age: 20
10 Course: Python
11 Total Marks: 225
12 Average Marks: 75.0
```

Try to write the program without looking at the previous examples.

Common Beginner Mistakes

Forgetting Quotation Marks

Incorrect:

```
1 print(Hello)
```

Correct:

```
1 print("Hello")
```

Using a Number at the Beginning of a Variable

Incorrect:

```
1 1name = "Rahul"
```

Correct:

```
1 name1 = "Rahul"
```

Using Spaces in Variable Names

Incorrect:

```
1 student name = "Rahul"
```

Correct:

```
1 student_name = "Rahul"
```

Forgetting to Convert Numerical Input

Incorrect:

```
1 a = input("Enter a number: ")
2 b = input("Enter another number: ")
3
4 print(a + b)
```

If the user enters 10 and 20, the result is:

```
1 1020
```

Correct:

```
1 a = int(input("Enter a number: "))
2 b = int(input("Enter another number: "))
3
4 print(a + b)
```

The result is:

```
1 30
```

Case Sensitivity

The following are different variables:

```
1 name = "Rahul"
2 Name = "Amit"
3 NAME = "John"
```

Python treats uppercase and lowercase letters as different.

Answers to Practice Exercises

The answers below are provided for checking your work after attempting the exercises yourself.

Answer 1: Personal Information

One possible answer is:

```
1 print("Name: Rahul")
2 print("Age: 20")
3 print("City: Guwahati")
4 print("Course: Python")
```

The actual values should be replaced with the student's own information.

Answer 2: Arithmetic

```
1 print(25 + 15)
2 print(50 - 17)
3 print(12 * 8)
4 print(100 / 4)
```

Output:

```
1 40
2 33
3 96
4 25.0
```

Answer 3: Variables

```
1 a = 25
2 b = 10
3
4 print(a + b)
5 print(a - b)
6 print(a * b)
7 print(a / b)
```

Output:

```
1 35
2 15
3 250
4 2.5
```

Answer 4: User Name

```
1 name = input("Enter your name: ")
2
3 print("Hello", name)
4 print("Welcome to Python programming.")
```

Answer 5: Age Calculator

```
1 age = int(input("Enter your age: "))
2
3 print("Next year you will be", age + 1)
```

Answer 6: Rectangle Area

```
1 length = float(input("Enter length: "))
2 width = float(input("Enter width: "))
3
4 area = length * width
5
6 print("Area:", area)
```

Answer 7: Shopping Total

```
1 price = float(input("Enter price: "))
2 quantity = int(input("Enter quantity: "))
3
4 total = price * quantity
5
6 print("Total cost:", total)
```

Answer 8: Square of a Number

```
1  number = float(input("Enter a number: "))
2
3  square = number ** 2
4
5  print("Square:", square)
```

Answer 9: Celsius to Fahrenheit

```
1  celsius = float(input("Enter temperature in Celsius: "))
2
3  fahrenheit = (9 / 5) * celsius + 32
4
5  print("Temperature in Fahrenheit:", fahrenheit)
```

Answer 10: Three Subject Marks

```
1  mark1 = float(input("Enter marks in Subject 1: "))
2  mark2 = float(input("Enter marks in Subject 2: "))
3  mark3 = float(input("Enter marks in Subject 3: "))
4
5  total = mark1 + mark2 + mark3
6  average = total / 3
7
8  print("Total:", total)
9  print("Average:", average)
```

Answer: Student Information Program

One possible solution is:

```
1  name = input("Enter student name: ")
2  age = int(input("Enter age: "))
3  course = input("Enter course: ")
4
5  mark1 = float(input("Enter marks in Subject 1: "))
6  mark2 = float(input("Enter marks in Subject 2: "))
```

```
7 mark3 = float(input("Enter marks in Subject 3: "))
8
9 total = mark1 + mark2 + mark3
10 average = total / 3
11
12 print()
13 print("Student Name:", name)
14 print("Age:", age)
15 print("Course:", course)
16 print("Total Marks:", total)
17 print("Average Marks:", average)
```

Final Practice Task

Without looking at the answers, create a program that asks the user for:

1. Product name
2. Product price
3. Quantity purchased
4. Customer name

The program should calculate the total amount and display a simple bill.

For example:

```
1 Enter customer name: Rahul
2 Enter product name: Notebook
3 Enter price: 50
4 Enter quantity: 4
5
6 Customer: Rahul
7 Product: Notebook
8 Price: 50.0
9 Quantity: 4
10 Total: 200.0
```


Answer: Final Practice Task

One possible solution is:

```
1  customer = input("Enter customer name: ")
2  product = input("Enter product name: ")
3  price = float(input("Enter price: "))
4  quantity = int(input("Enter quantity: "))
5
6  total = price * quantity
7
8  print()
9  print("Customer:", customer)
10 print("Product:", product)
11 print("Price:", price)
12 print("Quantity:", quantity)
13 print("Total:", total)
```

Module Review

Before moving to the next module, make sure you understand the following:

- What Python is
- What programming means
- How to run a Python program
- How `print()` works
- How to write comments
- What variables are
- How to name variables
- The difference between strings, integers, floats, and Booleans
- How arithmetic operators work
- How `input()` works
- Why `input()` returns a string

- How to use `int()` and `float()`
- How to create a simple interactive program

End of Python Module 1